

Boss of the SOC (BOTS) – Version 2

Dataset:

- The dataset used for the CTF is generated by members of Splunk's Security Speciality team - Dave Herrald, Ryan Kovar, Steve Brant, Jim Apger, John Stoner, Ken Westin, David Veuve and James Brodsky.
- It was originally created for **Splunk's Boss of the SOC competition**, a blue team capture-the-flag (CTF) event but later it made open-source.
- It used by many threat hunting, incident response and forensic investigators.
- There are **three versions** of this dataset.
- **BOTSv1** is focused on company named "Wayne Enterprises" where the attacks include web defacement, brute force, scanning and malware beaconing.
- **BOTSv2** is focused on company called "Frothly" a craft beer company, the attacks include spear phishing, data exfiltration, reconnaissance and persistence.
- **BOTSv3** is focused on the cloud-native attacks where compromised AWS credentials, advanced lateral movement and container threats are attacks.
- The full **BOTSv2** Dataset has a total size of 16.4GB.

Data Sourcetypes included

- access_combined
- activedirectory
- apache:error
- apache_error
- auditd
- bandwidth
- collectd
- cpu
- csp-violation

- df
- ess_content_importer
- hardware
- interfaces
- iostat
- lastlog
- linux:selinuxconfig
- linux_audit
- linux_secure
- ms:o365:management
- msad:nt6:health
- msad:nt6:siteinfo
- mysql:connection:stats
- mysql:database
- mysql:errorlog
- mysql:instance:stats
- mysql:server:stats
- mysql:status
- mysql:table_io_waits_summary_by_index_usage
- mysql:tablestatus
- mysql:transaction:details
- mysql:transaction:stats
- mysql:user
- mysql:variables
- mysqld-8
- netstat

- openports
- osquery_info
- osquery_results
- osquery_warning
- package
- pan:system
- pan:threat
- pan:traffic
- perfmon:cpu
- perfmon:logicaldisk
- perfmon:memory
- perfmon:network
- perfmon:network_interface
- perfmon:ntds
- perfmon:physicaldisk
- perfmon:process
- perfmon:processor
- perfmon:system
- powershell:scriptexecutionsummary
- protocol
- ps
- script:installedapps
- script:listeningports
- stream:arp
- stream:dhcp
- stream:dns

- stream:ftp
- stream:http
- stream:icmp
- stream:ip
- stream:irc
- stream:ldap
- stream:mysql
- stream:smb
- stream:smtp
- stream:tcp
- stream:udp
- suricata
- symantec:ep:agent:file
- symantec:ep:agt_system:file
- symantec:ep:behavior:file
- symantec:ep:packet:file
- symantec:ep:scan:file
- symantec:ep:scm_system:file
- symantec:ep:security:file
- symantec:ep:traffic:file
- syslog
- time
- top
- unix:listeningports
- unix:service
- unix:update

- unix:uptime
- unix:useraccounts
- unix:version
- userswithloginprivs
- vmstat
- web_ping
- weblogic_access_combined
- weblogic_stdout
- who
- windowsupdatelog
- wineventlog:application
- wineventlog:directory-service
- wineventlog:security
- wineventlog:system
- winhostmon
- winregistry
- xmlwineventlog:microsoft-windows-sysmon/operational

TryHackMe

I have used to solve the CTF using the **TryHackMe** platform using the attack box attached to the room called **Splunk2**.

Room Structure:

The room contains of series of questions which are named as 100 series through 400 series questions.

Catch-the-flag (CTF):

I have done 100 series questions and working on the other questions in the meantime.

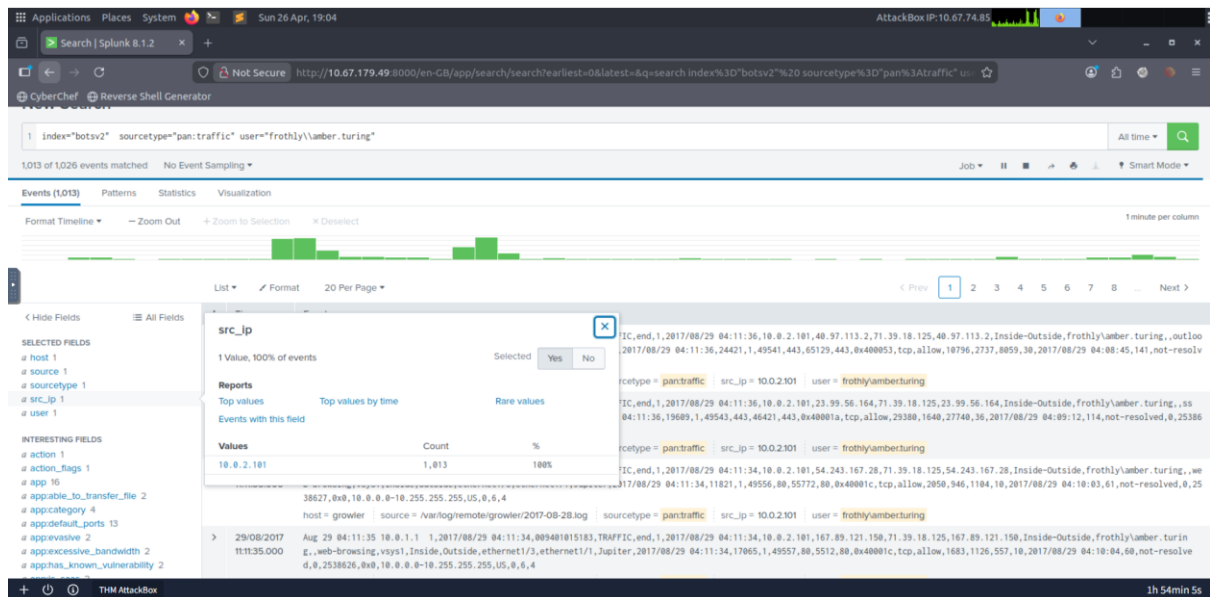
Attack Map:

At the end of the CTF, I included my attack map for my investigations.

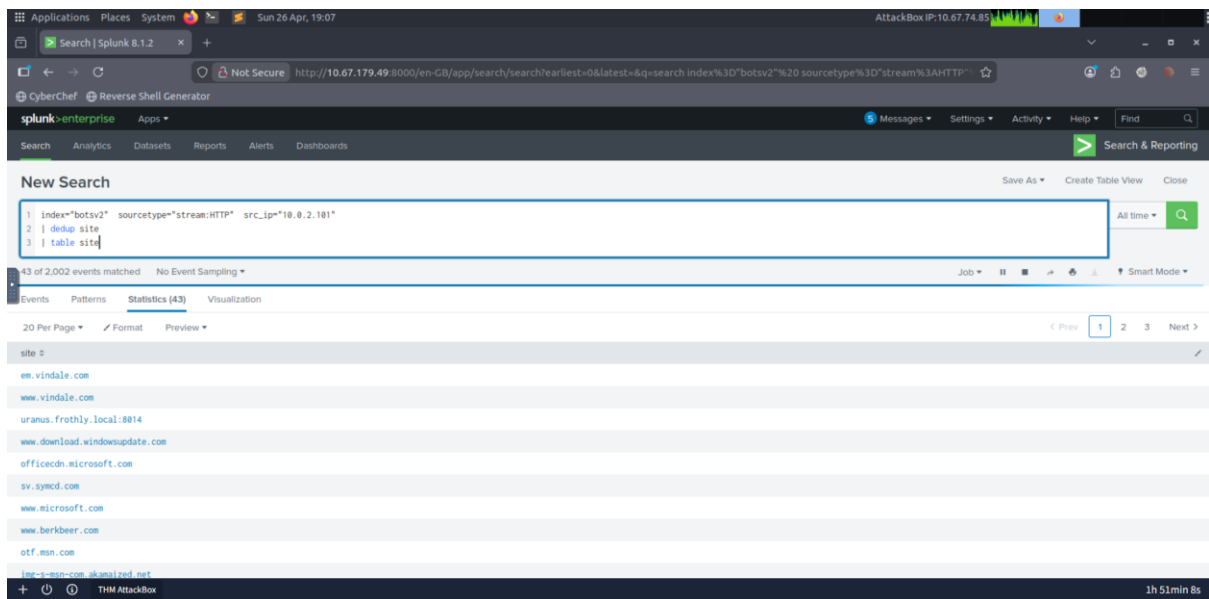
100 SERIES QUESTIONS

Question1: Amber Turing was hoping for Frothly to be acquired by a potential competitor which fell through but visited their website to find contact information for their executive team. What is the website domain that she visited?

I got the IP address of the amber using the sourcetype as pan:traffic and selecting user as amber.turing. Then src_ip is shown in selected fields.



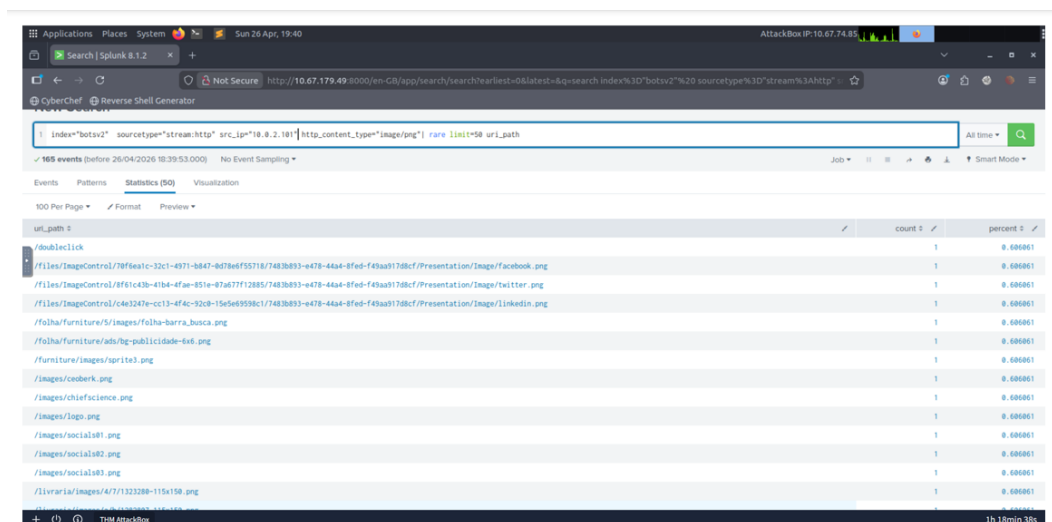
I used the `src_ip`, removed the duplicates of sites visited and created table format the non-duplicates of sites visited. As amber works in a beverage company and she is gonna search for her competitor and so, I guessed it would be berkbeer.



Answer: `www[.]berkbeer[.]com`

Question 2: Amber found the executive contact information and sent him an email. What image file displayed the executive's contact information? Answer example: `/path/image.ext`

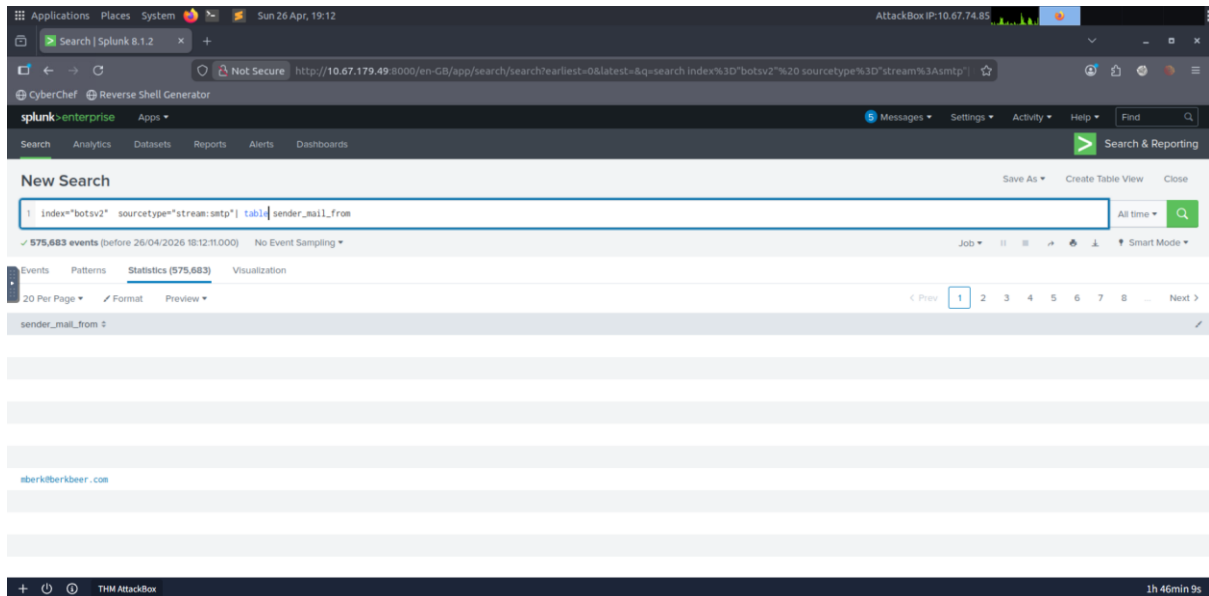
The known parameters are index, sourcetype and http content type and I used rare statistics on uri_path to get the results. As it mentioned as executive, I thought ceoberk might be the answer.



Answer: `/images/ceoberk.png`

Question 3: What is the CEO's name? Provide the first and last name.

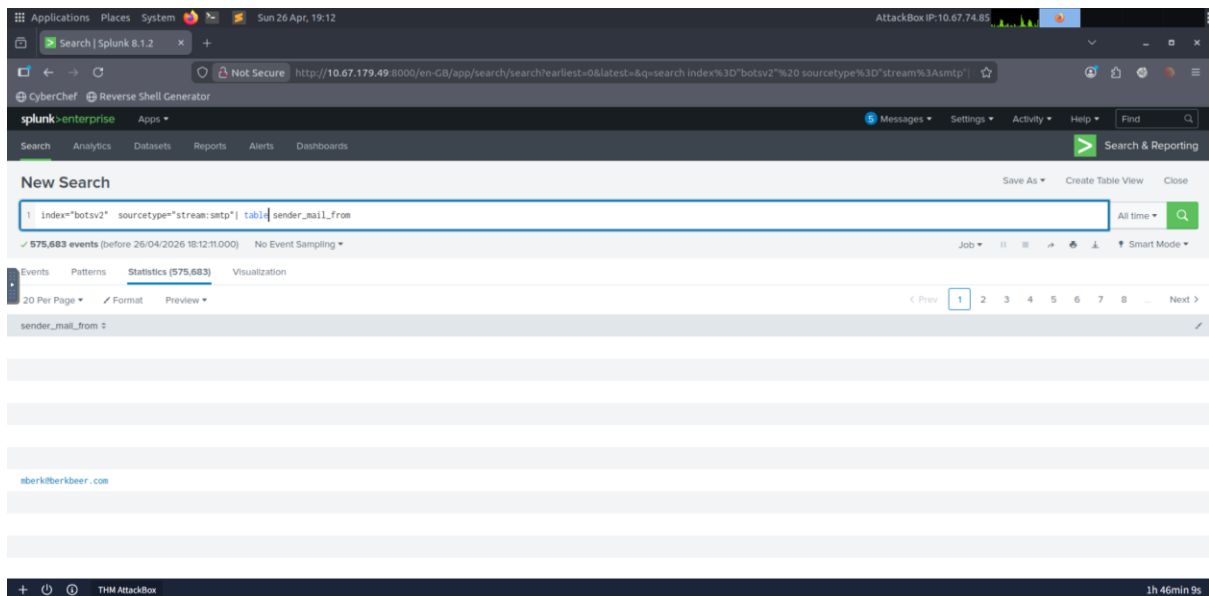
With clicking on the email id, it showed me list of events and got to know the full name of the CEO.



Answer: Martin Berk

Question 4: What is the CEO's email address?

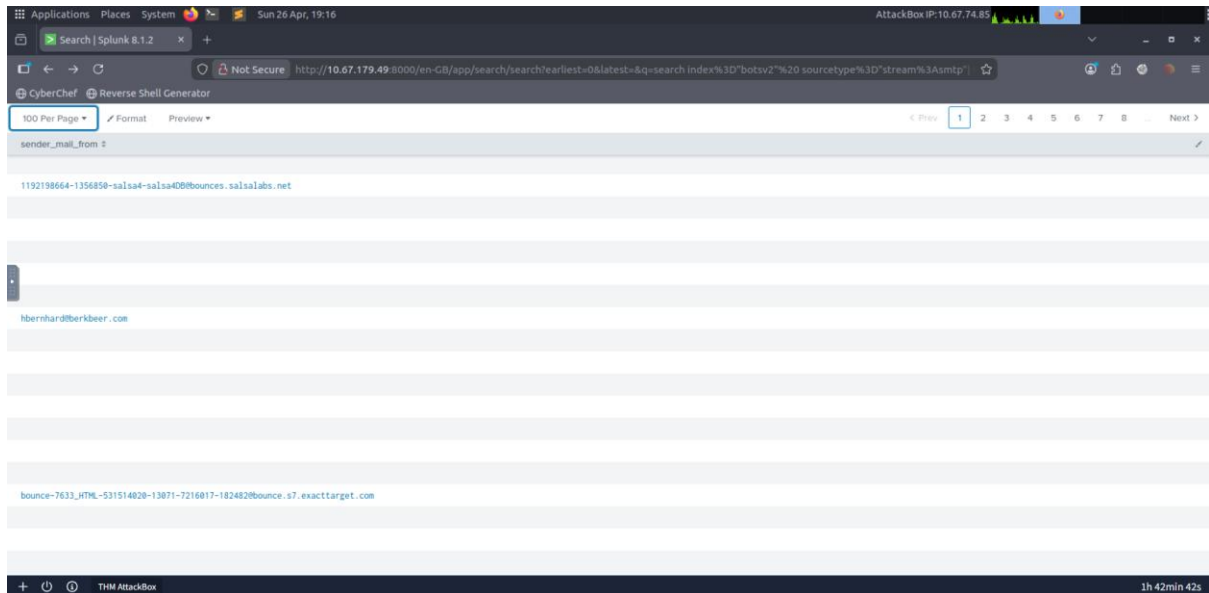
I have filtered using the field sender_table_from and made a table



Answer: mberk@berkbeer.com

Question 5: After the initial contact with the CEO, Amber contacted another employee at this competitor. What is that employee's email address?

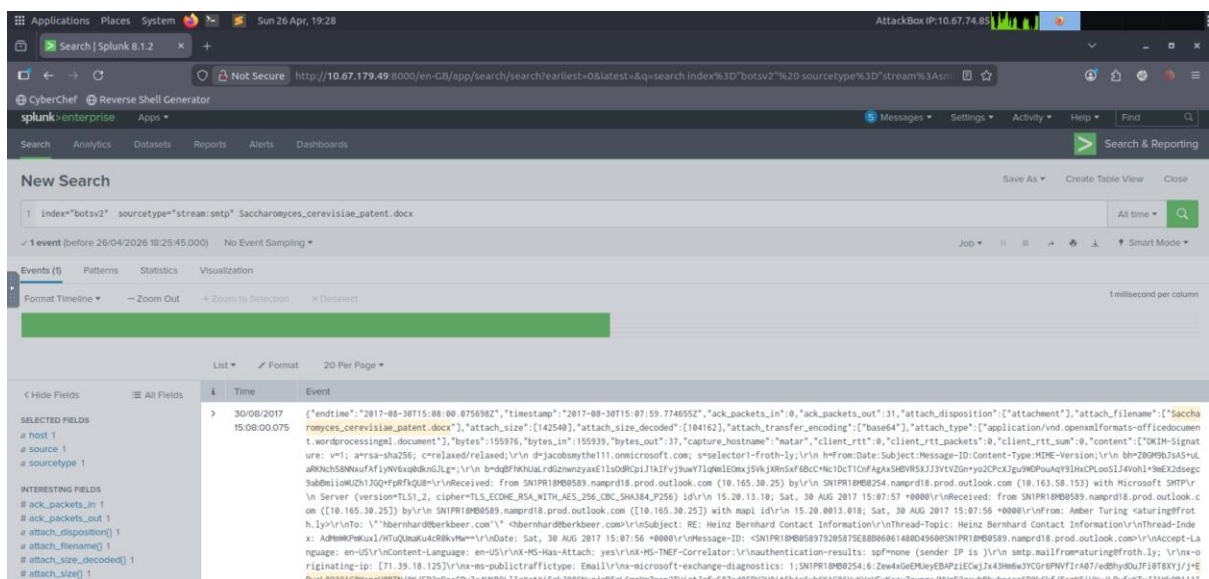
I have filtered using the field sender_table_from and made a table I found a email with domain berkbeer.



Answer: hbernhard@berkbeer.com

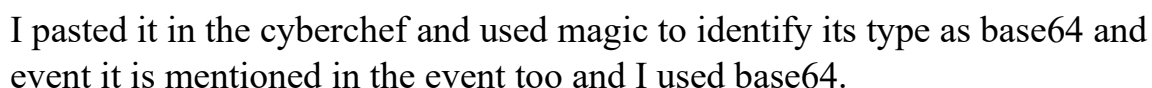
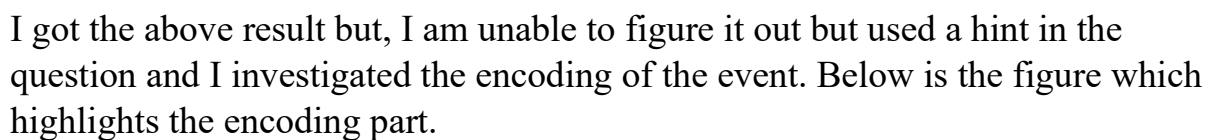
Question 6: What is the name of the file attachment that Amber sent to a contact at the competitor?

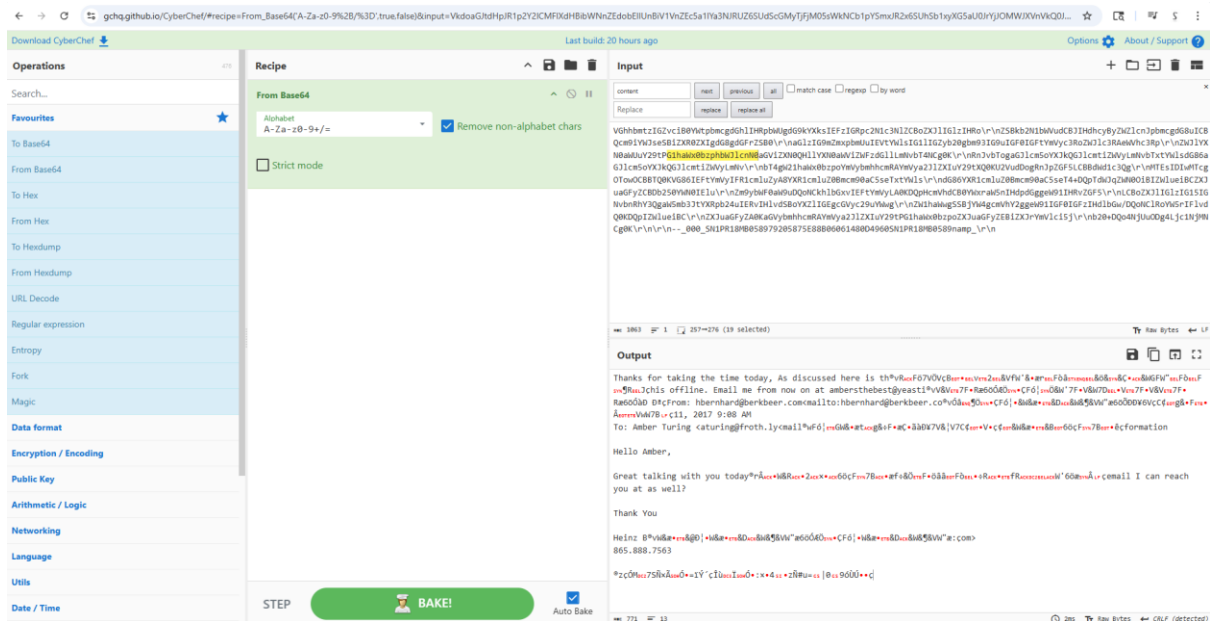
I used sourcetype as smtp and searched with keyword file



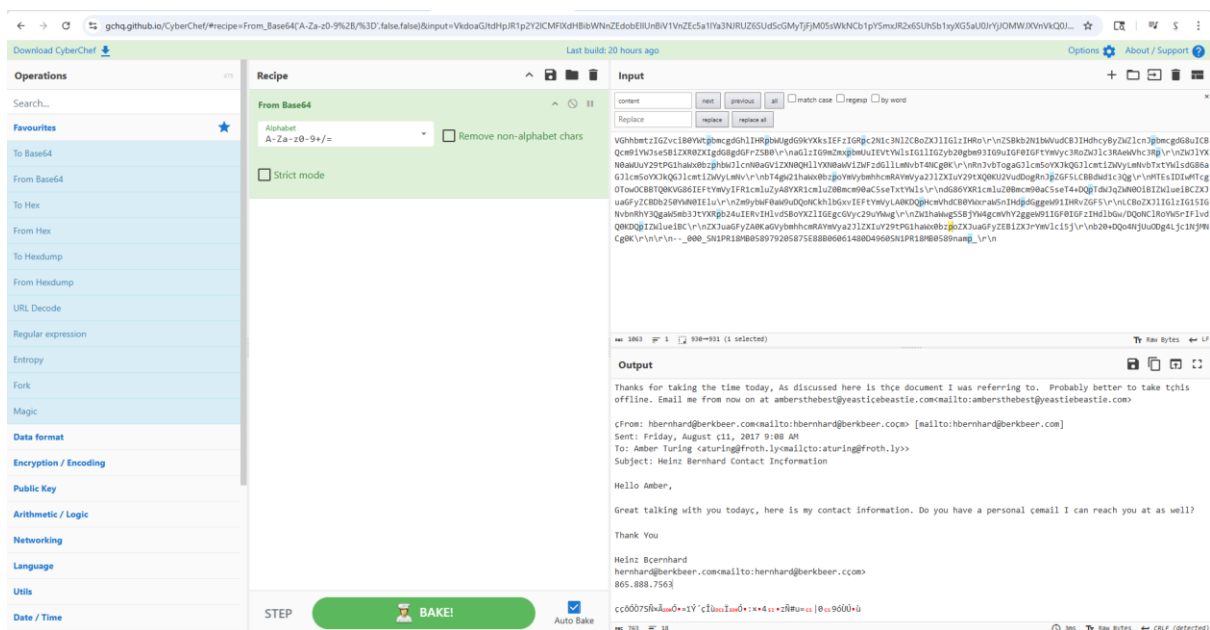
Answer: Saccharomyces_cerevisiae_patent.docx

I used file as filter and searched for it with sourcetype as smtp





I can get the personal email id of amber, but it is partially masked so, I unticked the option which said “Remove non-alphabet chars” then I am able to get the email id. Below is the picture which of the process



Answer: ambers@yeastiebeastie.com

Mapping the attack: Reconnaissance

Amber turning is the working professional of Frothly beer company. She visited the competitor company's website named **www[.]berkbeer[.]com**. She got the name as **Martin Berk** and found the email id as **mberk@berkbeer[.]com** at the website URL **/images/ceoberk.png** and in the email she asked the CEO for another person's email **hbernhard@berkbeer[.]com**. Later she sent a file called **Saccharomyces_cerevisiae_patent.docx** to them while mentioning her personal email address as **ambersthebest@yeastiebeastie[.]com**.